

DOCUMENTO DE ABERTURA E ENCERRAMENTO

Ordem de Serviço - Laboratorio 04B

Manutencao Continua do Hardening Linux

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Status: CONCLUIDA | Lynis HI=69 | security update aplicada

1. Identificacao

Empresa	AmazonTech
Unidade	Gerencia de Infraestrutura e Seguranca da Informacao
Servidor	WEB-01 (MachadoPC) - Debian 13.6
Tipo	Monitoramento, auditoria e manutencao continua
Data	06/08/2026
Equipe	Josias Bentes, Keven Coimbra, Margefson Barros e Nattan Lobato
Status da OS	CONCLUIDA

2. Necessidades x Tratamento

Situacao	Acao / Resultado	Status
Atualizacoes pendentes	1 pacote (libxpat1 security) - aplicado	Resolvido
Atualizacoes de seguranca	Debian-Security: libxpat1 2.8.2-1~deb13u1	Aplicado
Boletins / bases de seguranca	security.debian.org + debsecan (381 CVEs unicos)	Consultado
Vulnerabilidades conhecidas	debsecan; priorizar openssl/libc6 em ciclo	Avaliado
Integridade de arquivos	AIDE check + debsums -c (0 mismatches)	Validado
Auditoria de configuracao	Lynis HI=69; 0 warnings; 48 suggestions	Auditado
Evidencias	lab04b/evidencias + PDFs	Produzido

3. Evidencias principais

```
Terminal - Evidencia 2 - apt update

root@MachadoPC:~# apt-get update
Hit:1 http://ftp.debian.org/debian trixie-backports InRelease
Hit:2 http://deb.debian.org/debian trixie InRelease
Hit:3 http://security.debian.org/debian-security trixie-security InRelease
Hit:4 http://deb.debian.org/debian trixie-updates InRelease
Reading package lists...
```

Figura - Evidencia 2 - apt update

```
Terminal - Evidencia 4 - apt upgrade (security libxpat1)

root@MachadoPC:~# apt-get -s upgrade 2>&1 | head -60
Reading package lists...
Building dependency tree...
Reading state information...
Calculating upgrade...
The following packages will be upgraded:
  libxpat1
1 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
Inst libxpat1 [2.7.1-2] (2.8.2-1~deb13u1 Debian-Security:13/stable-security [amd64])
Conf libxpat1 (2.8.2-1~deb13u1 Debian-Security:13/stable-security [amd64])

root@MachadoPC:~# # Decisao: aplicar atualizacoes de seguranca no lab
root@MachadoPC:~# apt-get -y upgrade 2>&1 | tail -40
Reading package lists...
Building dependency tree...
Reading state information...
Calculating upgrade...
The following packages will be upgraded:
  libxpat1
1 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
Need to get 122 kB of archives.
After this operation, 23.6 kB of additional disk space will be used.
Get:1 http://security.debian.org/debian-security trixie-security/main amd64 libxpat1 amd64 2.8.2-1~deb13u1 [122 kB]
Fetched 122 kB in 0s (462 kB/s)
(Reading database ...
(Reading database ... 5%
(Reading database ... 10%
(Reading database ... 15%
(Reading database ... 20%
(Reading database ... 25%
(Reading database ... 30%
(Reading database ... 35%
(Reading database ... 40%
... (saida truncada) ...
```

Figura - Evidencia 4 - apt upgrade (security libxpat1)

```
Terminal - Evidencia 7 - Lynis audit (HI=69)

root@MachadoPC:~# lynis audit system
Lynis 3.1.4 - security audit

Hardening index : 69
Tests performed : 259
Plugins enabled : 1
Warnings       : 0 (Great, no warnings)
Suggestions    : 48

Components:
- Firewall      [V]
- Malware scanner [X]

Report: /var/log/lynis-report.dat
Log:    /var/log/lynis.log

Principais suggestions (prioridade AmazonTech):
1) AUTH-9286 / AUTH-9262 - politica de senha (PASS_MAX_DAYS / pam_pwquality)
2) DEB-0831 - needrestart apos upgrades
3) ACCT-9628 - auditd para trilha de auditoria
4) HRDN-7230 - malware scanner periodico
5) FINT-4402 - SHA256/SHA512 no AIDE
6) TIME-3104 - NTP client/daemon
7) MACF-6208 - validar status MAC (SELinux/AppArmor)
8) PKGS / apt-listchanges / apt-listbugs - visibilidade pre-upgrade

Conclusao: HI=69 estavel vs Lab03; 0 warnings; plano de melhoria continua via suggestions.
```

Figura - Evidencia 7 - Lynis audit (HI=69)

```
Terminal - Evidencia 8 - AIDE --check

root@MachadoPC:~# aide.wrapper --check
AIDE 0.19.1 | baseline: /var/lib/aide/aide.db | exit=7

Start timestamp: 2026-08-06 19:22:26 -0400 (AIDE 0.19.1)
AIDE found differences between database and filesystem!!
Added entries:
Removed entries:
Changed entries:
The attributes of the (uncompressed) database(s):
End timestamp: 2026-08-06 19:23:58 -0400 (run time: 1m 32s)

Entry /auditoria in databases has different attributes: +selinux
Entry /auditoria/parecer.txt in databases has different attributes: +selinux
Entry /bin in databases has different attributes: +selinux
Entry /etc in databases has different attributes: +selinux
Entry /etc/.pwd.lock in databases has different attributes: +selinux
Entry /etc/.updated in databases has different attributes: +selinux
Entry /etc/adduser.conf in databases has different attributes: +selinux
Entry /etc/aide in databases has different attributes: +selinux
Entry /etc/aide/aide.conf in databases has different attributes: +selinux
Entry /etc/aide/aide.conf.d in databases has different attributes: +selinux
Entry /etc/aide/aide.conf.d/01_aide_readcache in databases has different attributes: +selinux
Entry /etc/aide/aide.conf.d/10_aide_basenotations in databases has different attributes: +selinux
Entry /etc/aide/aide.conf.d/10_aide_bits in databases has different attributes: +selinux
Entry /etc/aide/aide.conf.d/10_aide_bootid in databases has different attributes: +selinux
Entry /etc/aide/aide.conf.d/10_aide_checksums in databases has different attributes: +selinux
Entry /etc/aide/aide.conf.d/10_aide_constants in databases has different attributes: +selinux
Entry /etc/aide/aide.conf.d/10_aide_dateformats in databases has different attributes: +selinux
Entry /etc/aide/aide.conf.d/10_aide_days in databases has different attributes: +selinux
Entry /etc/aide/aide.conf.d/10_aide_distribution in databases has different attributes: +selinux
Entry /etc/aide/aide.conf.d/10_aide_hardware in databases has different attributes: +selinux
Entry /etc/aide/aide.conf.d/10_aide_hostname in databases has different attributes: +selinux
Entry /etc/aide/aide.conf.d/10_aide_logext in databases has different attributes: +selinux
... (saida truncada) ...
```

Figura - Evidencia 8 - AIDE --check

```
Terminal - Evidencia 9 - debsums -c

root@MachadoPC:~# debsums -c 2>&1 | tee /tmp/debsums04b.txt | head -40

exit_code=0 (0=sem inconsistencias reportadas por -c)
linhas alteradas: 0
Nenhuma inconsistencia reportada por debsums -c (checksums OK ou silencio).

root@MachadoPC:~# debsums --help | head -15
debsums checks the MD5 sums of installed debian packages.

Usage: debsums [OPTIONS] [PACKAGE|DEB] ...

Options:
-a, --all                check configuration files (normally excluded)
-e, --config             check only configuration files
-c, --changed            report changed files (implies -s)
-l, --list-missing       list packages which don't have an md5sums file
-s, --silent             only report errors
-m, --md5sums=FILE       read list of deb checksums from FILE
-x, --report-mismatches  report errors and print the md5sums mismatch
-r, --root=DIR           root directory to check (default /)
-d, --admindir=DIR       dpkg admin directory (default /var/lib/dpkg)
-p, --deb-path=DIR[:DIR...] search path for debs
```

Figura - Evidencia 9 - debsums -c

4. Conformidade Politica AmazonTech

Diretriz		Atendida?
1. Servidor permanentemente atualizado		SIM
2. Prioridade a atualizacoes de seguranca		SIM
3. Boletins/aviso s consultados regularmente		SIM
4. Auditoria periodica da configuracao		SIM
5. Alteracoes de integridade investigadas		SIM
6. Evidencias tecnicas produzidas		SIM
7. Politica de Hardening revisada continuamente		SIM
8. Ferramentas complementam protecoes existentes		SIM
9. Minima indisponibilidade observada		SIM
10. Conclusoes subsidiam politica integrada (Aula 5)		SIM
Resultado	OS CONCLUIDA - ciclo de manutencao continua executado e documentado	
Documento tecnico	Lab04B_Manutencao_Continua_Hardening_PREENCHIDO.pdf	
Equipe	Josias Bentes, Keven Coimbra, Margefson Barros e Nattan Lobato	